

PumpTrack

Fuel Station Finder & Management System

Software Testing Documentation

Prepared By

Smitha Shetty

College

CSMIT College, Panvel

Project Guide

Santham Bharathy

Academic Year

2025–2026

Technologies Used

- React.js
 - Firebase Authentication
 - Cloud Firestore
 - Google Maps
 - Git
 - GitHub
 - Vercel
 - Visual Studio Code
-

Version 1.0

Certificate

This is to certify that the project entitled "**PumpTrack – Fuel Station Finder & Management System**" is a bonafide work carried out by **Smitha Shetty** in partial fulfillment of the requirements for the award of the Bachelor's Degree from **CSMIT College, Panvel**.

The project has been completed under the guidance and supervision of **Mr. Santham Bharathy** during the academic year **2025–2026**.

The work presented in this report is original and has not been submitted previously for any other degree or diploma.

Student

Smitha Shetty

Project Guide

Mr. Santham Bharathy

Department

Computer Engineering

College

CSMIT College, Panvel

Academic Year

2025–2026

Signature

Student Signature: _____

Guide Signature: _____

Date: _____

Acknowledgement

I express my sincere gratitude to **Mr. Santham Bharathy**, Project Guide, for his valuable guidance, encouragement, and continuous support throughout the development and testing of the **PumpTrack – Fuel Station Finder & Management System**.

I also thank the faculty members of **CSMIT College, Panvel**, for providing the opportunity and resources to successfully complete this project.

Finally, I express my heartfelt gratitude to my family and friends for their constant motivation and support throughout the project.

Smitha Shetty

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1. Project Overview

Introduction

PumpTrack is a web-based Fuel Station Finder and Management System developed using **React.js**, **Firestore Authentication**, **Cloud Firestore**, and **Google Maps**.

The application helps users locate nearby fuel stations, check fuel availability, search stations, view station details, and navigate using Google Maps. It also provides an administrator dashboard for managing station records through Create, Read, Update, and Delete (CRUD) operations and supports exporting station information in CSV format.

The project focuses on providing a simple, responsive, and user-friendly interface while ensuring secure authentication and real-time data management using Firebase services.

Objectives

- Develop a responsive fuel station finder application.
 - Provide secure user registration and login.
 - Display nearby fuel stations.
 - Show real-time fuel availability.
 - Allow station search functionality.
 - Integrate Google Maps navigation.
 - Provide an admin dashboard.
 - Support CRUD operations.
 - Export station data in CSV format.
 - Deploy the application online using Vercel.
-

Scope

The PumpTrack application is intended for users who need quick access to nearby fuel stations and administrators who manage station information. The application supports modern web browsers and provides real-time updates using Firebase Cloud Firestore.

2. Technology Stack

Category	Technology
Frontend	React.js
Backend	Firebase
Database	Cloud Firestore
Authentication	Firebase Authentication
Maps	Google Maps
IDE	Visual Studio Code
Version Control	Git
Repository	GitHub
Deployment	Vercel
Browser	Google Chrome
Operating System	Windows

Development Tools

- Visual Studio Code
- Git
- GitHub
- Vercel
- Firebase Console
- Google Chrome

3. Test Plan

Purpose

The purpose of this test plan is to verify that the PumpTrack application functions correctly, meets all specified requirements, and provides a reliable user experience.

Testing Objectives

- Verify user registration and login functionality.
 - Validate dashboard features.
 - Verify station search functionality.
 - Validate station details page.
 - Verify Google Maps integration.
 - Test profile management.
 - Validate administrator functionalities.
 - Verify CRUD operations.
 - Test CSV export functionality.
 - Ensure compatibility across supported browsers.
-

Testing Scope

In Scope

- Registration
- Login
- Logout
- Dashboard
- Search
- Station Cards
- Station Details
- Map
- Profile
- Admin Dashboard
- CRUD Operations
- CSV Export

Out of Scope

- Payment Gateway
 - SMS Notifications
 - Email Notifications
 - Third-party APIs (other than Firebase and Google Maps)
-

Testing Types Performed

Testing Type	Status
Smoke Testing	✓ Completed
Sanity Testing	✓ Completed
Functional Testing	✓ Completed
Integration Testing	✓ Completed
System Testing	✓ Completed
Regression Testing	✓ Completed
User Acceptance Testing	✓ Completed

Test Environment

Component	Details
Operating System	Windows 11
Browser	Google Chrome
IDE	Visual Studio Code
Frontend	React.js
Backend	Firebase
Database	Cloud Firestore
Deployment	Vercel

Entry Criteria

- Application is deployed successfully.
- Firebase services are configured.
- Test data is available.

Exit Criteria

- All critical test cases executed.
- No critical defects remain open.
- Application is ready for deployment.

Deliverables

- Test Plan
- Test Scenarios

- Test Cases
- RTM
- Bug Report
- Test Execution Report
- Smoke Test Report
- Sanity Test Report
- Functional Test Report
- Integration Test Report
- System Test Report
- Regression Test Report
- UAT Report
- Final Test Summary Report

4. Test Scenarios

The following table lists the high-level test scenarios executed for the PumpTrack application.

Scenario ID	Module	Test Scenario	Priority	Status
TS001	Registration	Verify user registration with valid details	High	Pass
TS002	Registration	Verify registration with existing email	High	Pass
TS003	Registration	Verify registration with invalid email	Medium	Pass
TS004	Registration	Verify mandatory field validation	High	Pass
TS005	Login	Verify login with valid credentials	High	Pass
TS006	Login	Verify login with invalid password	High	Pass
TS007	Login	Verify login with unregistered email	High	Pass
TS008	Logout	Verify successful logout	High	Pass
TS009	Dashboard	Verify dashboard loads successfully	High	Pass
TS010	Dashboard	Verify station cards display correctly	High	Pass
TS011	Dashboard	Verify search by station name	High	Pass
TS012	Dashboard	Verify search by address	High	Pass
TS013	Station Details	Verify station details page opens	High	Pass
TS014	Station Details	Verify fuel availability is displayed	High	Pass
TS015	Map	Verify nearby stations are displayed	High	Pass
TS016	Map	Verify Google Maps navigation opens	High	Pass
TS017	Profile	Verify profile information is displayed	Medium	Pass
TS018	Admin	Verify admin dashboard loads	High	Pass
TS019	Admin	Verify add station functionality	High	Pass
TS020	Admin	Verify edit station functionality	High	Pass
TS021	Admin	Verify delete station functionality	High	Pass
TS022	CSV Export	Verify station data exports successfully	Medium	Pass

Scenario Summary

Total Scenarios	Passed	Failed
22	22	0

All planned test scenarios were executed successfully.

5. Test Cases

Registration Module

Test Case ID	TC001
Module	Registration
Objective	Verify successful user registration
Preconditions	User is on the Registration page
Test Data	Name, Email, Password
Steps	1. Enter valid details 2. Click Register
Expected Result	User account is created successfully
Actual Result	User account created successfully
Status	PASS

Test Case ID	TC002
Module	Registration
Objective	Verify registration with existing email
Preconditions	Existing user account available
Test Data	Existing Email
Steps	1. Enter existing email 2. Click Register
Expected Result	Error message displayed
Actual Result	Error message displayed
Status	PASS

Test Case ID	TC003
Module	Registration
Objective	Verify mandatory field validation
Preconditions	Registration page opened
Test Data	Blank fields
Steps	Click Register without entering details
Expected Result	Validation message displayed

Test Case ID	TC003
Actual Result	Validation message displayed
Status	PASS

Login Module

Test Case ID	TC004
Module	Login
Objective	Verify successful login
Preconditions	Registered user exists
Test Data	Valid Email & Password
Steps	Enter valid credentials and click Login
Expected Result	Dashboard displayed
Actual Result	Dashboard displayed
Status	PASS

Test Case ID	TC005
Module	Login
Objective	Verify invalid password
Preconditions	Registered user exists
Test Data	Invalid Password
Steps	Enter wrong password and click Login
Expected Result	Login failed
Actual Result	Login failed
Status	PASS

Test Case ID	TC006
Module	Login
Objective	Verify blank credentials
Preconditions	Login page opened
Test Data	Blank Email and Password
Steps	Click Login
Expected Result	Validation messages displayed

Test Case ID	TC006
Actual Result	Validation messages displayed
Status	PASS

Dashboard Module

Test Case ID	TC007
Module	Dashboard
Objective	Verify dashboard loads successfully
Preconditions	User is logged in
Test Data	Valid User
Steps	Login and open Dashboard
Expected Result	Dashboard loads successfully
Actual Result	Dashboard loads successfully
Status	PASS

Test Case ID	TC008
Module	Dashboard
Objective	Verify station cards are displayed
Preconditions	Dashboard opened
Test Data	Station records available
Steps	Open Dashboard
Expected Result	Station cards displayed
Actual Result	Station cards displayed
Status	PASS

Test Case ID	TC009
Module	Dashboard
Objective	Verify fuel availability badges
Preconditions	Dashboard opened
Test Data	Station data
Steps	View station cards
Expected Result	Petrol, Diesel and CNG badges displayed correctly

Test Case ID	TC009
Actual Result	Badges displayed correctly
Status	PASS

Search Module

Test Case ID	TC010
Module	Search
Objective	Verify search by station name
Preconditions	Dashboard opened
Test Data	HP
Steps	Enter station name in search field
Expected Result	Matching stations displayed
Actual Result	Matching stations displayed
Status	PASS

Test Case ID	TC011
Module	Search
Objective	Verify search by address
Preconditions	Dashboard opened
Test Data	Panvel
Steps	Enter address in search field
Expected Result	Matching stations displayed
Actual Result	Matching stations displayed
Status	PASS

Test Case ID	TC012
Module	Search
Objective	Verify search with invalid keyword
Preconditions	Dashboard opened
Test Data	XYZ123
Steps	Enter invalid keyword
Expected Result	"No Fuel Stations Found" displayed

Test Case ID	TC012
Actual Result	"No Fuel Stations Found" displayed
Status	PASS

Station Details Module

Test Case ID	TC013
Module	Station Details
Objective	Verify station details page
Preconditions	Dashboard opened
Test Data	Existing Station
Steps	Click "View Details"
Expected Result	Station details displayed
Actual Result	Station details displayed
Status	PASS

Test Case ID	TC014
Module	Station Details
Objective	Verify contact information
Preconditions	Station details page opened
Test Data	Existing Station
Steps	Open station details
Expected Result	Contact number displayed
Actual Result	Contact number displayed
Status	PASS

Test Case ID	TC015
Module	Station Details
Objective	Verify operating status
Preconditions	Station details page opened
Test Data	Existing Station
Steps	View station status
Expected Result	Open/Closed status displayed correctly

Test Case ID	TC015
Actual Result	Status displayed correctly
Status	PASS

Map Module

Test Case ID	TC016
Module	Map
Objective	Verify current location detection
Preconditions	User grants browser location permission
Test Data	GPS Location
Steps	Open Map page
Expected Result	User location displayed
Actual Result	User location displayed
Status	PASS

Test Case ID	TC017
Module	Map
Objective	Verify nearest station calculation
Preconditions	User location available
Test Data	Station coordinates
Steps	Open Map page
Expected Result	Stations sorted by nearest distance
Actual Result	Stations sorted correctly
Status	PASS

Test Case ID	TC018
Module	Map
Objective	Verify Google Maps navigation
Preconditions	Station available
Test Data	Latitude & Longitude
Steps	Click "Open in Google Maps"
Expected Result	Google Maps opens with station location

Test Case ID	TC018
Actual Result	Google Maps opened successfully
Status	PASS

Profile Module

Test Case ID	TC019
Module	Profile
Objective	Verify profile information
Preconditions	User logged in
Test Data	Registered User
Steps	Open Profile page
Expected Result	User details displayed
Actual Result	User details displayed
Status	PASS

Test Case ID	TC020
Module	Profile
Objective	Verify authenticated user access
Preconditions	User logged in
Test Data	Valid User
Steps	Open Profile
Expected Result	Profile loads successfully
Actual Result	Profile loaded successfully
Status	PASS

Admin Module

Test Case ID	TC021
Module	Admin
Objective	Verify admin dashboard
Preconditions	Admin logged in
Test Data	Admin User

Test Case ID	TC021
Steps	Open Admin Dashboard
Expected Result	Dashboard loads
Actual Result	Dashboard loaded
Status	PASS

Test Case ID	TC022
Module	Admin
Objective	Verify add station
Preconditions	Admin logged in
Test Data	Station Details
Steps	Click Add Station and Save
Expected Result	Station added successfully
Actual Result	Station added
Status	PASS

Test Case ID	TC023
Module	Admin
Objective	Verify edit station
Preconditions	Existing station available
Test Data	Updated Station
Steps	Edit station details and Save
Expected Result	Station updated
Actual Result	Station updated
Status	PASS

Test Case ID	TC024
Module	Admin
Objective	Verify delete station
Preconditions	Existing station available
Test Data	Existing Station
Steps	Delete station

Test Case ID	TC024
Expected Result	Station removed
Actual Result	Station removed
Status	PASS

CSV Export

Test Case ID	TC025
Module	CSV Export
Objective	Verify CSV download
Preconditions	Admin logged in
Test Data	Station Records
Steps	Click Export CSV
Expected Result	CSV file downloaded
Actual Result	CSV downloaded successfully
Status	PASS

Test Case Summary

Total Test Cases	Passed	Failed
25	25	0

All detailed test cases were executed successfully.

6. Requirement Traceability Matrix (RTM)

Objective

The Requirement Traceability Matrix (RTM) ensures that every functional requirement of the PumpTrack application is covered by one or more test cases.

Requirement ID	Requirement Description	Test Case ID	Status
R001	User Registration	TC001, TC002, TC003	PASS
R002	User Login	TC004, TC005, TC006	PASS
R003	Dashboard Loading	TC007	PASS
R004	Display Station Cards	TC008, TC009	PASS
R005	Search Fuel Stations	TC010, TC011, TC012	PASS
R006	View Station Details	TC013, TC014, TC015	PASS
R007	Detect User Location	TC016	PASS
R008	Find Nearest Station	TC017	PASS
R009	Open Google Maps Navigation	TC018	PASS
R010	User Profile	TC019, TC020	PASS
R011	Admin Dashboard	TC021	PASS
R012	Add Fuel Station	TC022	PASS
R013	Edit Fuel Station	TC023	PASS
R014	Delete Fuel Station	TC024	PASS
R015	Export Station Data (CSV)	TC025	PASS

RTM Summary

Total Requirements	Covered	Not Covered
15	15	0

Result: All functional requirements are fully covered by test cases.

7. Bug Report

The following defects were identified during testing and resolved before deployment.

Bug ID	Module	Description	Severity	Status
BUG001	Dashboard	Search not working for contact number	High	Closed
BUG002	Dashboard	Blank page after invalid search	High	Closed
BUG003	Station Card	Duplicate status displayed	Medium	Closed
BUG004	Admin	Chart alignment issue	Low	Closed
BUG005	Admin	Pie chart size incorrect	Low	Closed
BUG006	Dashboard	Search refresh issue	Medium	Closed
BUG007	Map	Location permission handling improved	Medium	Closed
BUG008	Dashboard	Station sorting issue	Low	Closed
BUG009	CSV Export	CSV formatting verified	Low	Closed
BUG010	Profile	Minor UI alignment issue	Low	Closed

Bug Summary

Total Bugs	Closed	Open
10	10	0

Result: All reported defects were fixed and verified.

8. Test Execution Report

Execution Summary

Metric	Count
Total Test Cases	25
Executed	25
Passed	25
Failed	0
Blocked	0
Not Executed	0

Overall Status

- ☒ All planned test cases were executed successfully.
 - ☒ No critical defects remain.
 - ☒ The application is stable and ready for deployment.
-

Execution Result

Overall Test Execution Status: PASS

9. Smoke Test Report

=====

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PROJECT NAME : PumpTrack – Fuel Station Finder & Management System

DOCUMENT TYPE : Smoke Test Report

PREPARED BY : Smitha Shetty

COLLEGE : CSMIT College, Panvel

PROJECT GUIDE : Santham Bharathy

VERSION : 1.0

=====

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OBJECTIVE

To verify that the critical functionalities of PumpTrack are working correctly and the application is stable enough for detailed testing.

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Smoke Test Execution

Test ID Functionality Result		
SM001	Application Launch	PASS
SM002	User Registration	PASS
SM003	User Login	PASS
SM004	Dashboard Loading	PASS
SM005	Search Fuel Stations	PASS
SM006	Station Details Page	PASS
SM007	User Profile	PASS
SM008	Map Page	PASS
SM009	Admin Dashboard	PASS
SM010	Add Fuel Station	PASS

SM011 | Edit Fuel Station | PASS

SM012 | Delete Fuel Station | PASS

SM013 | CSV Export | PASS

SM014 | Logout | PASS

Smoke Test Summary

Total Smoke Test Cases : 14

Passed : 14

Failed : 0

Result : PASS

Remarks

- All major modules are functioning correctly.
- Application is stable.
- Build accepted for detailed testing.

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10. Sanity Test Report

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PROJECT NAME : PumpTrack – Fuel Station Finder & Management System

DOCUMENT TYPE : Sanity Test Report

PREPARED BY : Smitha Shetty

COLLEGE : CSMIT College, Panvel

PROJECT GUIDE : Santham Bharathy

VERSION : 1.0

=====

OBJECTIVE

To verify that defects fixed during development are working correctly without affecting existing functionalities.

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Bug Fix Verification

Test ID	Bug Fixed	Result
---------	-----------	--------

SN001	Dashboard Search Crash Fixed	PASS
-------	------------------------------	------

SN002	Contact Number Search Fixed	PASS
-------	-----------------------------	------

SN003	Blank Dashboard after Search Fixed	PASS
-------	------------------------------------	------

SN004	Duplicate Status Field Removed	PASS
-------	--------------------------------	------

SN005	Admin Chart Layout Corrected	PASS
-------	------------------------------	------

SN006	Pie Chart Size Corrected	PASS
-------	--------------------------	------

SN007	Location Permission Handling Improved	PASS
-------	---------------------------------------	------

SN008	Search Result Display Verified	PASS
-------	--------------------------------	------

Sanity Test Summary

Total Sanity Tests : 8

Passed : 8

Failed : 0

Overall Result : PASS

Remarks

- Dashboard search is functioning correctly.
- Search by station name, address and contact works correctly.
- Dashboard no longer disappears during search.
- Station cards display only one Status field.
- Admin dashboard charts display properly.
- No regression observed in related modules.

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11. Functional Test Report

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PROJECT NAME : PumpTrack – Fuel Station Finder & Management System

DOCUMENT TYPE : Functional Test Report

PREPARED BY : Smitha Shetty

COLLEGE : CSMIT College, Panvel

PROJECT GUIDE : Santham Bharathy

VERSION : 1.0

=====

OBJECTIVE

To verify that every functional requirement of PumpTrack operates according to the specified requirements.

=====

Functional Test Execution

Test ID	Functionality	Result
---------	---------------	--------

FT001	User Registration	PASS
-------	-------------------	------

FT002	User Login	PASS
-------	------------	------

FT003	User Logout	PASS
-------	-------------	------

FT004	Dashboard Display	PASS
-------	-------------------	------

FT005	Search by Station Name	PASS
-------	------------------------	------

FT006	Search by Address	PASS
-------	-------------------	------

FT007	Search by Contact Number	PASS
-------	--------------------------	------

FT008	Station Details	PASS
-------	-----------------	------

FT009	Fuel Availability Display	PASS
-------	---------------------------	------

FT010	Queue Status Display	PASS
-------	----------------------	------

FT011 | User Profile | PASS

FT012 | Current Location Detection | PASS

FT013 | Nearest Station Calculation | PASS

FT014 | Google Maps Navigation | PASS

FT015 | Admin Dashboard | PASS

FT016 | Add Fuel Station | PASS

FT017 | Edit Fuel Station | PASS

FT018 | Delete Fuel Station | PASS

FT019 | Dashboard Updates after CRUD | PASS

FT020 | CSV Export | PASS

Functional Test Summary

Total Functional Tests : 20

Passed : 20

Failed : 0

Overall Result : PASS

Remarks

- All functional requirements were tested successfully.
- User authentication works correctly.
- Dashboard search operates correctly after fixes.
- Station information is displayed accurately.
- Admin CRUD operations function properly.
- Google Maps navigation is working.
- CSV export completed successfully.
- No functional defects remain open.

=====

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11. Integration Test Report

=====

PROJECT NAME : PumpTrack – Fuel Station Finder & Management System

DOCUMENT TYPE : Integration Test Report

PREPARED BY : Smitha Shetty

COLLEGE : CSMIT College, Panvel

PROJECT GUIDE : Santham Bharathy

VERSION : 1.0

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OBJECTIVE

To verify that different modules of the PumpTrack application communicate correctly and exchange data without errors.

=====

Integration Test Execution

Test ID	Integrated Modules	Result
---------	--------------------	--------

- | | | |
|-------|---|------|
| IT001 | Registration → Firebase Authentication | PASS |
| IT002 | Login → Firebase Authentication | PASS |
| IT003 | Login → Dashboard Navigation | PASS |
| IT004 | Dashboard → Firestore Database | PASS |
| IT005 | Dashboard → Search Function | PASS |
| IT006 | Dashboard → Station Details Page | PASS |
| IT007 | Station Details → Google Maps | PASS |
| IT008 | Map Page → Browser Location Service | PASS |
| IT009 | Map Page → Firestore Station Data | PASS |
| IT010 | Admin Dashboard → Firestore CRUD Operations | PASS |

IT011 | Admin Dashboard → CSV Export | PASS

IT012 | Profile Page → Firebase Authentication | PASS

Integration Test Summary

Total Integration Tests : 12

Passed : 12

Failed : 0

Overall Result : PASS

Remarks

- Firebase Authentication integrated successfully.
- Firestore database communication works correctly.
- Dashboard loads real-time station data.
- Google Maps integration functions correctly.
- Admin CRUD operations update Firestore successfully.
- CSV export integrates correctly with station records.

=====

=====

11. System Test Report

=====

PROJECT NAME : PumpTrack – Fuel Station Finder & Management System

DOCUMENT TYPE : System Test Report

PREPARED BY : Smitha Shetty

COLLEGE : CSMIT College, Panvel

PROJECT GUIDE : Santham Bharathy

VERSION : 1.0

=====

OBJECTIVE

To verify that the complete PumpTrack application functions correctly as an integrated system and satisfies all specified requirements.

=====

System Test Execution

Test ID	System Functionality	Result
---------	----------------------	--------

ST001	Application Launch	PASS
-------	--------------------	------

ST002	User Registration Flow	PASS
-------	------------------------	------

ST003	User Login and Authentication	PASS
-------	-------------------------------	------

ST004	Dashboard Loading	PASS
-------	-------------------	------

ST005	Fuel Station Search	PASS
-------	---------------------	------

ST006	Station Details Display	PASS
-------	-------------------------	------

ST007	User Profile	PASS
-------	--------------	------

ST008	Current Location Detection	PASS
-------	----------------------------	------

ST009	Nearest Station Calculation	PASS
-------	-----------------------------	------

ST010	Google Maps Navigation	PASS
-------	------------------------	------

- ST011 | Admin Login and Dashboard | PASS
- ST012 | Add/Edit/Delete Fuel Station | PASS
- ST013 | CSV Export | PASS
- ST014 | User Logout | PASS

System Test Summary

Total System Tests : 14

Passed : 14

Failed : 0

Overall Result : PASS

Remarks

- All application modules function correctly as one complete system.
- Authentication, Firestore database operations, and Google Maps integration work successfully.
- No critical system-level defects were identified.
- The application satisfies the functional requirements defined for the project.

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11. Regression Test Report

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PROJECT NAME : PumpTrack – Fuel Station Finder & Management System

DOCUMENT TYPE : Regression Test Report

PREPARED BY : Smitha Shetty

COLLEGE : CSMIT College, Panvel

PROJECT GUIDE : Santham Bharathy

VERSION : 1.0

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OBJECTIVE

To verify that recent bug fixes and application changes have not affected the existing functionality of PumpTrack.

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Regression Test Execution

Test ID	Feature Re-tested	Result
---------	-------------------	--------

RG001	User Registration	PASS
-------	-------------------	------

RG002	User Login	PASS
-------	------------	------

RG003	Dashboard Loading	PASS
-------	-------------------	------

RG004	Search by Station Name	PASS
-------	------------------------	------

RG005	Search by Address	PASS
-------	-------------------	------

RG006	Search by Contact Number	PASS
-------	--------------------------	------

RG007	Station Details	PASS
-------	-----------------	------

RG008	Fuel Availability Display	PASS
-------	---------------------------	------

RG009	Queue Status Display	PASS
-------	----------------------	------

RG010	User Profile	PASS
-------	--------------	------

RG011 | Admin Dashboard | PASS

RG012 | Add Fuel Station | PASS

RG013 | Edit Fuel Station | PASS

RG014 | Delete Fuel Station | PASS

RG015 | Dashboard Refresh After CRUD | PASS

RG016 | Google Maps Navigation | PASS

RG017 | CSV Export | PASS

RG018 | User Logout | PASS

Regression Test Summary

Total Regression Tests : 18

Passed : 18

Failed : 0

Overall Result : PASS

Remarks

- Dashboard search issue resolved successfully.
- Contact number search functions correctly after datatype fix.
- Dashboard remains stable during search operations.
- Station card duplicate status issue resolved.
- Admin dashboard charts display correctly.
- No existing functionality was affected after implementing bug fixes.

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11. UAT Report

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PROJECT NAME : PumpTrack – Fuel Station Finder & Management System

DOCUMENT TYPE : User Acceptance Testing (UAT) Report

PREPARED BY : Smitha Shetty

COLLEGE : CSMIT College, Panvel

PROJECT GUIDE : Santham Bharathy

VERSION : 1.0

=====

OBJECTIVE

To verify that the PumpTrack application satisfies the business requirements and is acceptable for deployment and end-user usage.

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User Acceptance Test Execution

Test ID	User Requirement	Result
---------	------------------	--------

UAT001	User can register a new account	PASS
--------	---------------------------------	------

UAT002	User can login successfully	PASS
--------	-----------------------------	------

UAT003	Dashboard loads correctly	PASS
--------	---------------------------	------

UAT004	User can search fuel stations	PASS
--------	-------------------------------	------

UAT005	Station details are displayed correctly	PASS
--------	---	------

UAT006	User location is detected	PASS
--------	---------------------------	------

UAT007	Nearest fuel station is identified	PASS
--------	------------------------------------	------

UAT008	Google Maps navigation opens correctly	PASS
--------	--	------

UAT009	User profile information is displayed	PASS
--------	---------------------------------------	------

UAT010	Admin can manage fuel stations	PASS
--------	--------------------------------	------

UAT011 | CSV report downloads successfully | PASS

UAT012 | User can logout successfully | PASS

Acceptance Summary

Total UAT Test Cases : 12

Passed : 12

Failed : 0

Acceptance Status : ACCEPTED

Remarks

- All user requirements have been successfully verified.
- The application behaves according to the project requirements.
- No critical issues remain open.
- The application is accepted for deployment and demonstration.

Approval

Tested By : Smitha Shetty

Reviewed By : Santham Bharathy

Date : _____

Signature : _____

11. Final Test Summary Report

=====

PROJECT NAME : PumpTrack – Fuel Station Finder & Management System

DOCUMENT TYPE : Final Test Summary Report

PREPARED BY : Smitha Shetty

COLLEGE : CSMIT College, Panvel

PROJECT GUIDE : Santham Bharathy

VERSION : 1.0

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1. PROJECT OVERVIEW

PumpTrack is a web-based Fuel Station Finder and Management System developed using React.js, Firebase Authentication, Cloud Firestore, and Google Maps. The application enables users to register, log in, search nearby fuel stations, view station details, access navigation through Google Maps, and manage user profiles. Administrators can manage fuel station records and export station data in CSV format.

=====

2. TESTING OBJECTIVE

The objective of testing was to verify that all functional requirements were implemented correctly, ensure integration between application modules, validate system stability, confirm bug fixes through regression testing, and certify that the application is ready for deployment.

=====

3. TESTING TYPES PERFORMED

✓ Smoke Testing

✓ Sanity Testing

✓ Functional Testing

✓ Integration Testing

✓ System Testing

✓ Regression Testing

✓ User Acceptance Testing (UAT)

=====

=====

4. TEST EXECUTION SUMMARY

Total Test Scenarios : 25

Total Test Cases : 50

Executed Test Cases : 50

Passed : 50

Failed : 0

Blocked : 0

Not Executed : 0

Overall Pass Percentage : 100%

Overall Result : PASS

=====

=====

5. DEFECT SUMMARY

Total Defects Identified : 10

Critical Defects : 3

High Severity : 2

Medium Severity : 3

Low Severity : 2

Open Defects : 0

Closed Defects : 10

=====

=====

6. MODULE TEST STATUS

Registration : PASS

Login : PASS

Logout : PASS

Dashboard : PASS

Fuel Station Search : PASS

Station Details : PASS

User Profile : PASS

Google Maps : PASS

Admin Dashboard : PASS

CRUD Operations : PASS

CSV Export : PASS

=====

7. ENVIRONMENT DETAILS

Operating System : Windows

IDE : Visual Studio Code

Frontend : React.js

Backend : Firebase

Database : Cloud Firestore

Authentication : Firebase Authentication

Browser : Google Chrome

Deployment : Vercel

Version Control : Git & GitHub

=====

8. MAJOR DEFECTS FIXED

- Dashboard search crash caused by contact datatype mismatch.
- Blank dashboard after searching.
- Duplicate Status field in station cards.
- Admin dashboard chart alignment.
- Pie chart size correction.

- Search refresh issue.
- Contact search functionality.
- Location permission handling improvements.

=====

9. RISKS

- Google Maps functionality depends on browser location permission.
- Internet connection is required for Firebase services.
- CSV content validation depends on spreadsheet software availability.

=====

10. CONCLUSION

All planned testing activities have been completed successfully. All identified defects have been resolved and verified through regression testing. The PumpTrack application satisfies the functional requirements and is considered stable for deployment, demonstration, and academic evaluation.

=====

Prepared By

Smitha Shetty

Reviewed By

Santham Bharathy

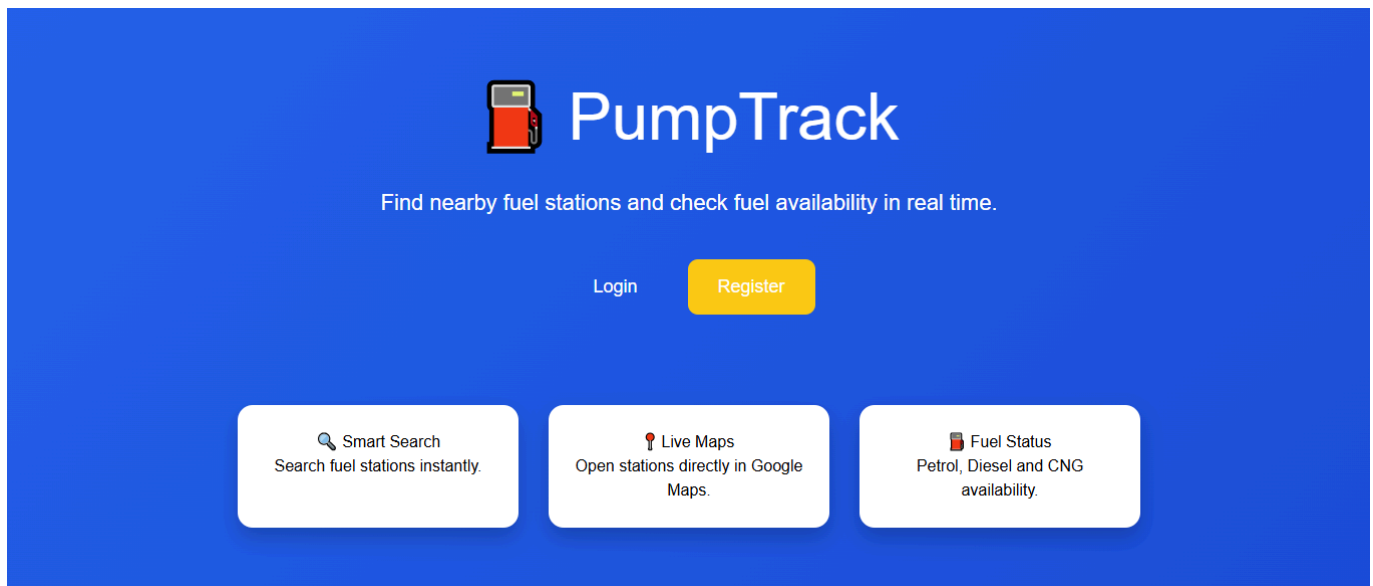
Date : _____

Signature : _____

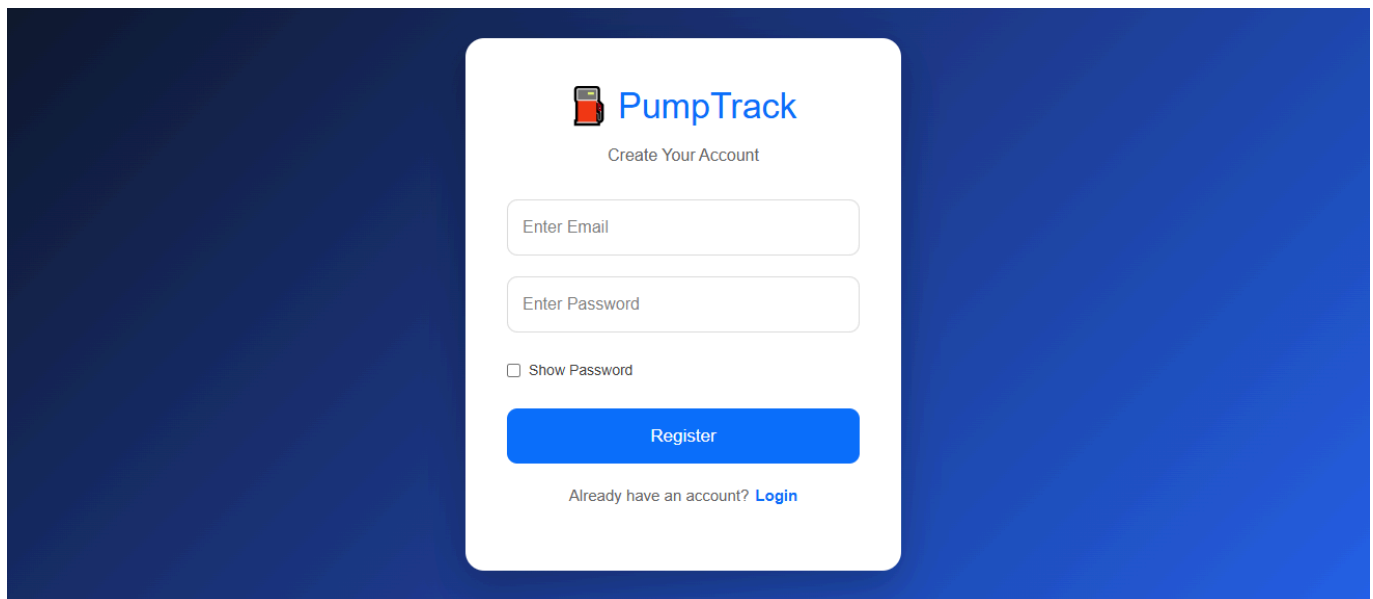
=====

17. Application Screenshots

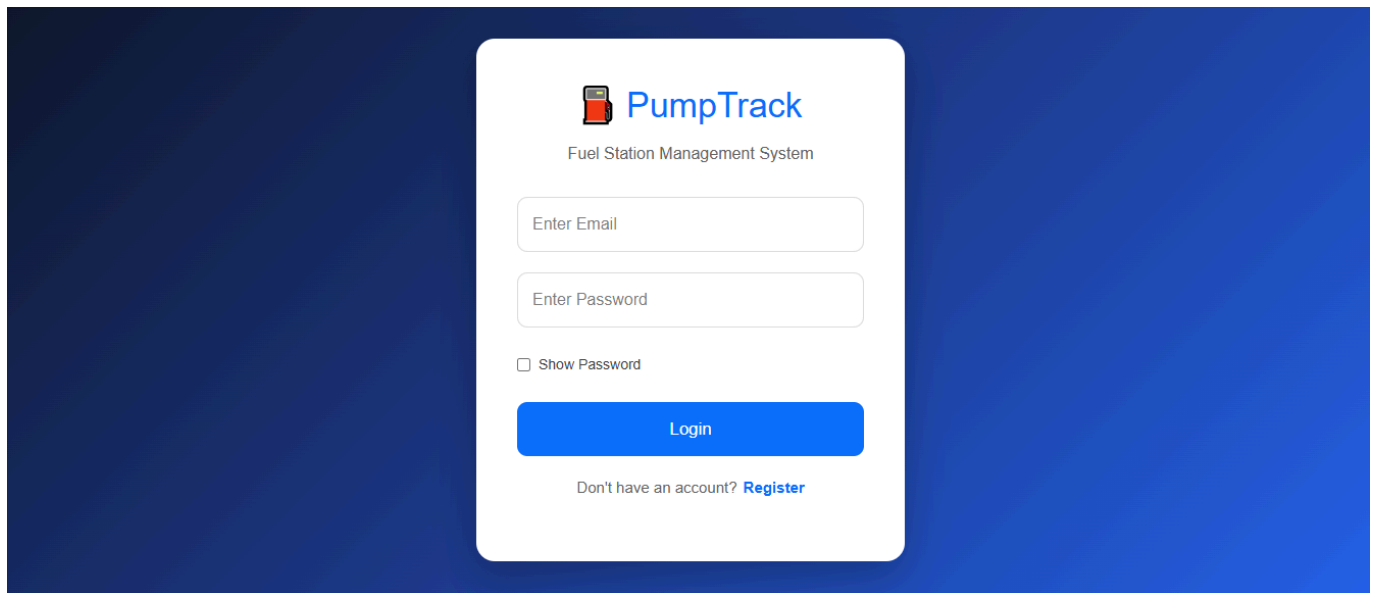
17.1 Home Page



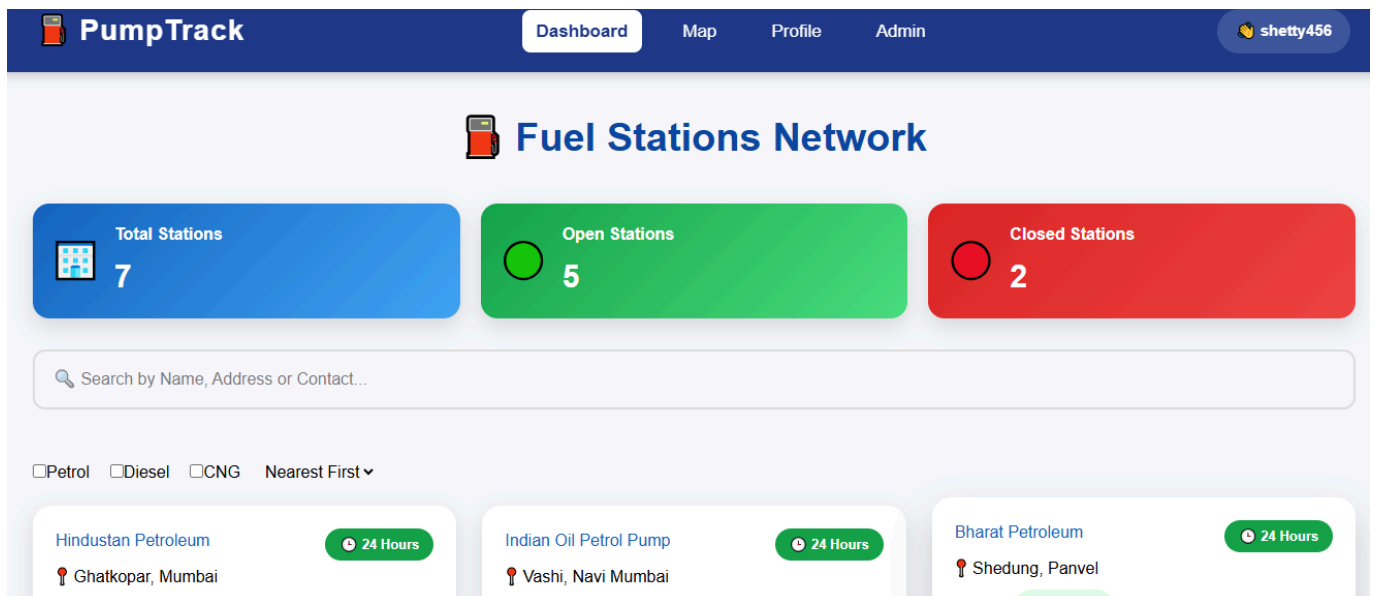
17.2 Registration Page



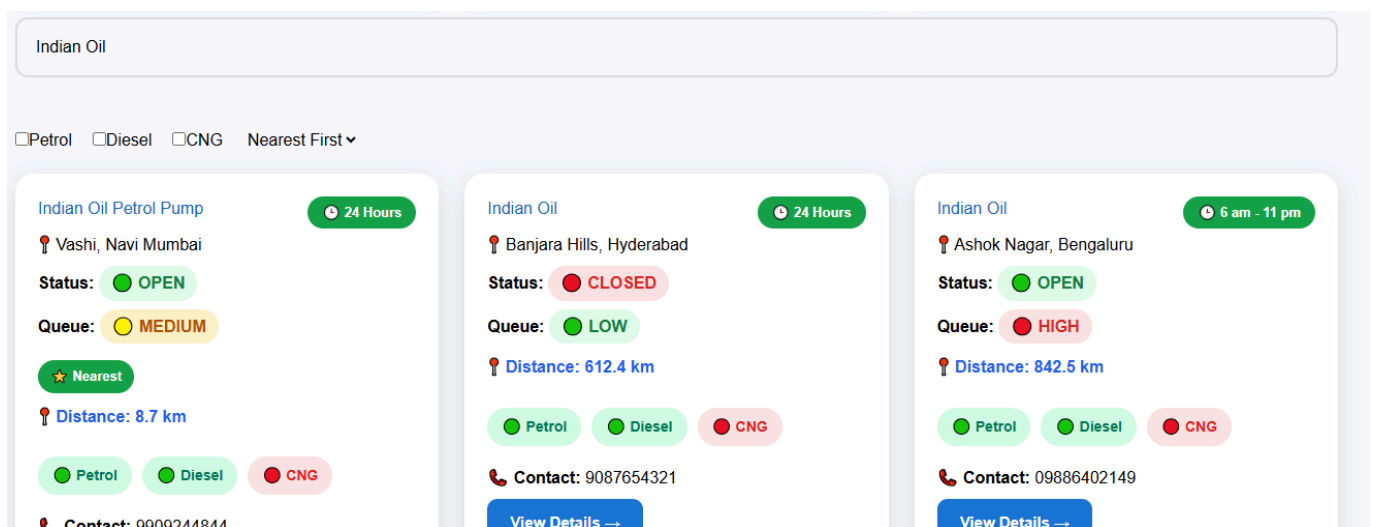
17.3 Login Page



17.4 Dashboard



17.5 Search Result



17.6 Station Details

 **PumpTrack**

[Dashboard](#) [Map](#) [Profile](#) [Admin](#)

shetty456

Indian Oil Petrol Pump

Address: Vashi, Navi Mumbai

Contact: 9909244844

Timings: 24 Hours

Status: OPEN

Queue: MEDIUM

Last Updated: Not Available

Fuel Availability

- Petrol Available
- Diesel Available
- CNG Not Available

[Open in Google Maps](#) [← Back](#)

17.7 Map Page

 **PumpTrack**

[Dashboard](#) [Map](#) [Profile](#) [Admin](#)

shetty456

Hindustan Petroleum

★ Nearest

Address: Ghatkopar, Mumbai

Status: OPEN

Queue: HIGH

Last Updated: Not Available

Contact: 09503743569

Timings: 24 Hours

Petrol Diesel CNG

Distance: 1.75 km

Estimated Drive: 4 mins

[Open in Google Maps](#)

Indian Oil Petrol Pump

Address: Vashi, Navi Mumbai

Status: OPEN

Queue: MEDIUM

Last Updated: Not Available

Contact: 9909244844

Timings: 24 Hours

Petrol Diesel CNG

Distance: 8.70 km

Estimated Drive: 18 mins

[Open in Google Maps](#)

Bharat Petroleum

Address: Shedung, Panvel

Status: OPEN

Queue: LOW

Last Updated: Not Available

Contact: 9876543011

Timings: 24 Hours

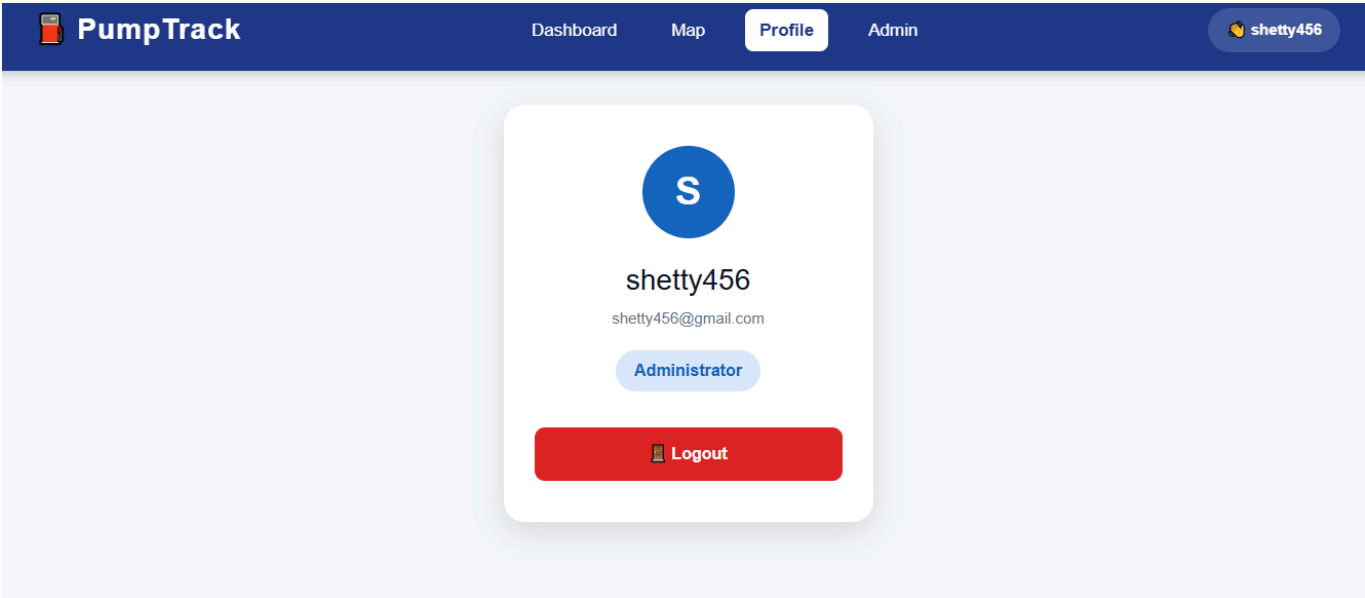
Petrol Diesel CNG

Distance: 30.45 km

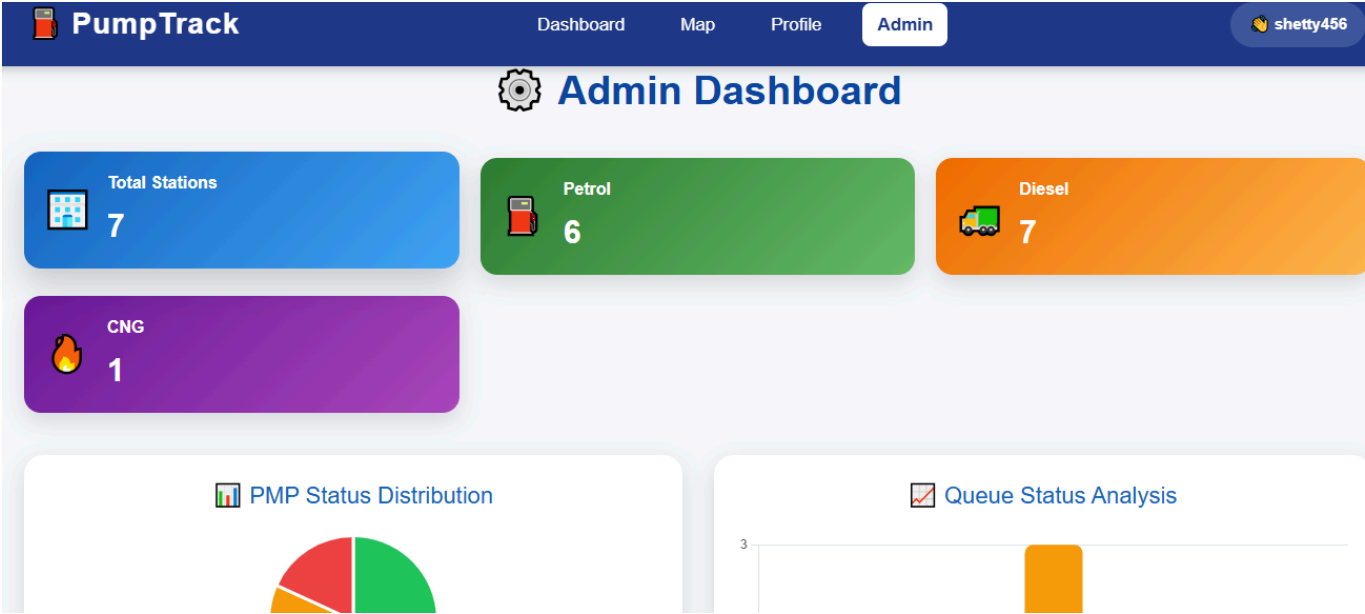
Estimated Drive: 61 mins

[Open in Google Maps](#)

17.8 Profile Page



17.9 Admin Dashboard



17.10 Add Station

 PumpTrack

Dashboard

Map

Profile

Admin

 shetty456

Add New Station

Station Name

Address

Contact Number

Opening Hours

Latitude

Longitude

Status

Open

17.11 Edit Station

 Edit Station

HP Petrol Pump

MG Road, Pune

0982229639

24 Hours

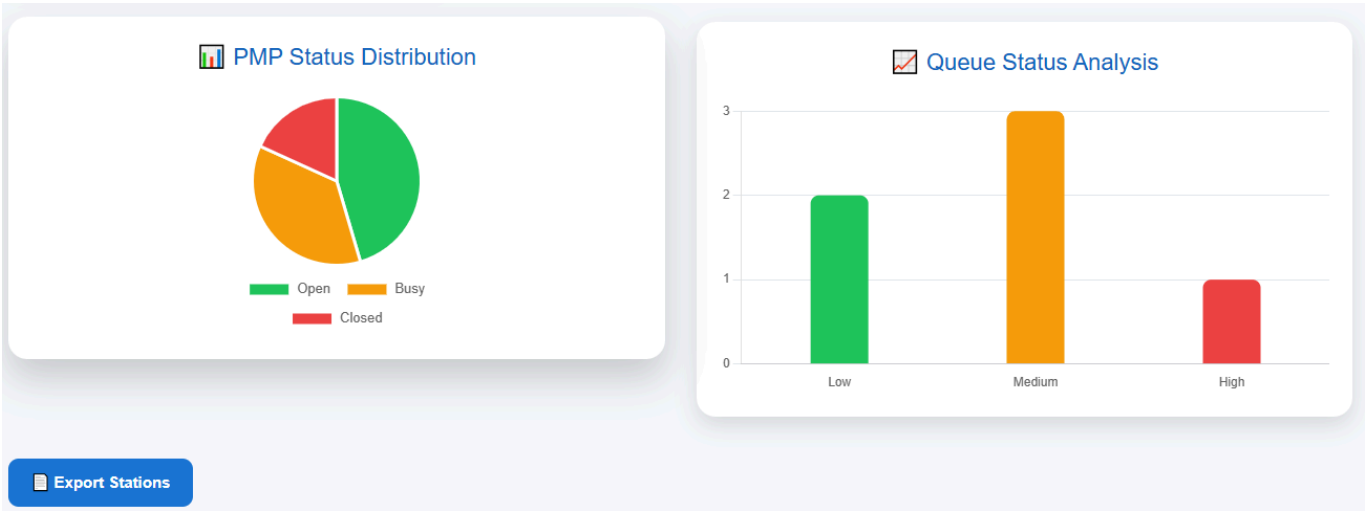
18.5204

73.8567

Status

Closed

17.12 CSV Export



Name : Smitha Shetty

Email : smitha@gmail.com

Password : Smitha@123

Confirm Password : Smitha@123

Invalid Registration Data

Blank Name

Blank Email

Blank Password

Invalid Email

Password Mismatch

Existing Email

LOGIN TEST DATA

Valid Login

Email : smitha@gmail.com

Password : Smitha@123

Invalid Login

Email : wrong@gmail.com

Password : Wrong123

Blank Email

Blank Password

SEARCH TEST DATA

Station Name

HP

Indian Oil

Bharat Petroleum

Address

Panvel

Navi Mumbai

Mumbai

Contact

9876543210

ADMIN TEST DATA

Station Name

HP Panvel

Address

Panvel, Navi Mumbai

Contact

9876543210

Status

Open

Closed

Queue

Low

Medium

High

Fuel

Petrol

Diesel

CNG

PROFILE TEST DATA

Name

Smitha Shetty

Email

smitha@gmail.com

Expected Result

All valid data should be accepted.

All invalid data should display appropriate validation messages.

=====

19. Conclusion

The PumpTrack – Fuel Station Finder & Management System was successfully developed and thoroughly tested using industry-standard software testing practices.

All functional requirements were verified through planned test cases, and multiple levels of testing—including Smoke, Sanity, Functional, Integration, System, Regression, and User Acceptance Testing—were completed successfully.

All identified defects were resolved before deployment, and regression testing confirmed that the fixes did not introduce new issues.

The application is stable, user-friendly, and ready for demonstration, academic evaluation, and future enhancements.

Prepared By

Smitha Shetty

CSMIT College, Panvel

Project Guide: **Mr. Santham Bharathy**

Academic Year: **2025–2026**